

Applied Regression II

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HW 5

In Homework 4, we fit Cox proportional hazards regression models to study risk factors for the breast cancer recurrence with the `hwdata1.csv`: Model 1 with the hormone therapy group as the only covariate; Model 2 with hormone therapy groups, age, menopause status, tumor size, and number of nodes as covariates; Model 3 with hormone therapy groups, age, menopause status, tumor size, number of nodes, and the interaction between hormone therapy groups and number of nodes as covariates.

Note: For further exploration on the individual models, please refer to the bottom of the homework (after problem three and before the code section)

QUESTION 1:

Compare Model 1 and Model 2. State the null and alternative hypothesis, test statistic (state which test was used), p-value, degrees of freedom, and conclusion.

- **Null Hypothesis:** $H_0 : \beta_{age} = \beta_{menopausal} = \beta_{size} = \beta_{nodes} = 0$
- **Alternative Hypothesis:** $H_A : \text{At least one } \beta \neq 0$
- **Test Statistic:** $G \sim \chi^2_{m2-m1}$
- **Test: Partial likelihood ratio test:** $G = 2[l_{m2}(\hat{\beta}) - l_{m1}(\hat{\beta})]$
 - It is partial because Cox model is semiparametric
- **Define alpha:** $\alpha = 0.05$
- **Degrees of Freedom:** 4
 - It is the number of different groups between the models

Model Comparison: Full Output

Model	Log Likelihood	LRT Test Statistic (G)	Df	p-value
Model 1	-1783.694	NA	NA	NA
Model 2	-1755.155	57.0778	4	0

Key Values from Partial Likelihood Ratio Test

LRT Test Statistic (G)	Degrees of Freedom	p-value
57.0778	4	0

Conclusion: Based on the partial likelihood ratio test, at a predefined $\alpha = 0.05$ we get $G = 57.0778 \sim \chi_4^2$ with a $p < 0.001$ (R rounds to 0). As $p < 0.05$, we reject the null hypothesis that age, menopausal status, tumor size, and number of involved nodes all together have no effect on the hazard of recurrence ($H_0 : \beta_{age} = \beta_{menopausal} = \beta_{size} = \beta_{nodes} = 0$). Further more, these results suggest that Model 2 provides a significantly better fit than Model 1: the additional covariates collectively improve the model.

QUESTION 2:

Compare Model 2 and Model 3. State the null and alternative hypothesis, test statistic (state which test was used), p-value, degrees of freedom, and conclusion.

- **Null Hypothesis:** $H_0 : \beta_{nodes*hormone} = 0$
- **Alternative Hypothesis:** $H_A : \beta_{nodes*hormone} \neq 0$
- **Test Statistic:** $G \sim \chi_{m3-m2}^2$
- **Test: Partial likelihood ratio test:** $G = 2[l_{m3}(\hat{\beta}) - l_{m2}(\hat{\beta})]$
 - It is partial because Cox model is semiparametric
- **Define alpha:** $\alpha = 0.05$
- **Degrees of Freedom:** 1
 - It is the number of different groups between the models

Table 3: Full Model Output

Model	Log Likelihood	LRT Test Statistic (G)	Df	p-value
Model 2	-1755.155	NA	NA	NA
Model 3	-1752.312	5.6862	1	0.0171

LRT Test Statistic (G)	Degrees of Freedom	p-value
5.686	1	0.017

Conclusion: Based on the partial likelihood ratio test, at a predefined $\alpha = 0.05$ we get $G = 5.686 \sim \chi_1^2$ with $p = 0.017$. As $p < 0.05$, we reject the null hypothesis that the association between hormone therapy and recurrence does not change as the number of nodes changes ($H_0 : \beta_{nodes*hormone} = 0$). Further more, these results suggest that Model 3 provides a significantly better fit than Model 2: the addition of the interaction term improves model fit.

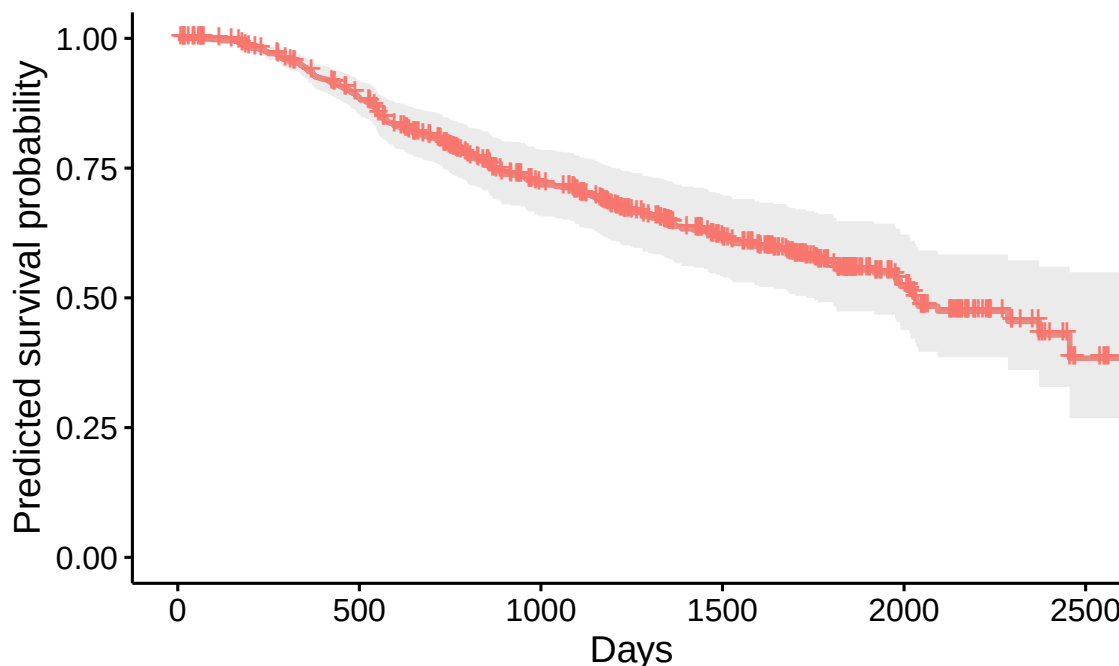
QUESTION 3:

Based on Model 3, draw the survival curve for a patient under hormone therapy, at age 53, being menopausal, having tumor of size 25mm, and having 3 nodes involved. What is the probability that this patient survives more than three years (1095 days)? Provide code and relevant output.

Part 1: Plot survival function for the specified patient

The survival function $S(t | X)$ is the survival probability over time for one specific patient (or $P(T > t | X)$).

Survival Curve of Specified Patient



Part 2: Calculate the probability of patient survival beyond 3 years (1095 days)

$$P(T > 1095 | X) = S(1095 | X)$$

Survival Statistics at 3 years for Specified Patient

Table 5: Predicted 3-Year Survival Probability (Model 3)

Time	Survival Probability	Std. Error	Lower 95% CI	Upper 95% CI
1095	0.705	0.034	0.642	0.774

Interpreting output: $S(1095 | X) = 0.705$: The estimated probability that this patient survives more than 1095 days (3 years) is 70.5%.

lower 95% CI: 0.642 Upper 95% CI 0.774: We are 95% confident that the true predicted probability this patient survives more than 3 years lies between 64.2% and 77.4%.

n.risk = 333: The number of subjects *still at risk* (event-free and under observation) just before $t = 1095$ in the original dataset. Value used to estimate the baseline survival.

n.event = 224: The number of observed recurrence events that occurred around 1095 days in the risk set.

Individual Model Explanations:

Notes on Data: Based on the dataset, variable “hormone” equaled 1 for patients under hormone therapy and 2 for patients not under hormone therapy. Those not under hormone therapy were considered the reference group. Additionally, variable “menopause” equaled 1 for participants who were experiencing menopause and 2 for those who were not experiencing menopause. The group of participants who were not experiencing menopause were the reference group for this variable.

Model 1 Model 1 is a Cox proportional hazards model including hormone therapy group as the only covariate

- **Covariate:** Hormone Therapy Group
- **Model form:** $\lambda(t | hormone) = \lambda_0 e^{\beta_1 * hormone}$
- **Fitted model ($\hat{\beta}$ rounded to three decimals):** $\lambda(t | hormone) = \lambda_0 e^{0.364 * hormone}$

Model 1 output: exponentiated coefficient

Variable	Hazard Ratio	Std Error	Z statistic	p-value	Lower 95% CI	Upper 95% CI
Hormone: Yes	1.439	0.125	2.911	0.004	1.126	1.839

The corresponding hazard ratio for hormone therapy is $\exp(0.364) = 1.439$. These results suggest that patients in hormone therapy have 1.439 times the hazard of recurrence compared to those not under hormone therapy (95%CI: 1.126, 1.839). That the 95% confidence interval does not include the null value of 1 (no difference in hazard between hormone therapy group and reference group) and that the p-value=0.004, suggests there is statistically significant evidence at $\alpha = 0.05$ that hormone therapy is associated with an increased hazard of recurrence.

Model 2 Model 2 extends Model 1 by adjusting for additional covariates.

- **Covariates:** Hormone therapy groups, Age, Menopause status, Tumor size, and Number of nodes
- **Model form:** $\lambda(t | X) = \lambda_0 e^{\beta_1 * hormone + \beta_2 * age + \beta_3 * menopausal + \beta_4 * size + \beta_5 * nodes}$
- **Fitted model ($\hat{\beta}$ rounded to three decimals):**

$$\lambda(t | X) = \lambda_0 e^{0.381 * hormone - 0.014 * age - 0.353 * menopausal + 0.008 * size + 0.052 * nodes}$$

Model 2 output: exponentiated coefficients

Variable	Hazard Ratio	Std Error	Z statistic	p-value	Lower 95% CI	Upper 95% CI
Hormone: Yes	1.464	0.128	2.970	0.003	1.138	1.882
Age (years)	0.987	0.009	-1.506	0.132	0.969	1.004
Menopausal: Yes	0.702	0.179	-1.977	0.048	0.495	0.997
Tumor size (mm)	1.008	0.004	2.013	0.044	1.000	1.016
Nodes involved	1.053	0.007	7.109	0.000	1.038	1.069

After adjustment for age, menopausal status, tumor size, and number of involved lymph nodes, the hazard ratio for hormone therapy is $\exp(0.381) = 1.464$ (95% CI: 1.138–1.882, p-value = 0.003). That the hazard ratio for hormone therapy remains similar between Model 1 (HR = 1.439) and Model 2 (HR = 1.464) suggests that the added covariates (age, menopausal status, tumor size, and number of involved nodes) do not substantially confound the association between hormone therapy and recurrence. However, the number of nodes (variable = nodes) appears to have the greatest association to hazard of occurrence. For every additional node the hazard increases by 5.3% (HR: 1.053, p-value << 0.05 after adjusting for the other covariates).

Model 3 Model 3 further extends Model 2 by including an interaction term between hormone therapy and number of nodes

- **Covariates:** Hormone therapy groups, Age, Menopause status, Tumor size, Number of nodes, and the Interaction between hormone therapy groups and number of nodes
- **Model form:** $\lambda(t | X) = \lambda_0 e^{\beta_1 * hormone + \beta_2 * age + \beta_3 * menopausal + \beta_4 * size + \beta_5 * nodes + \beta_6 * nodes * hormone}$
- **Fitted model ($\hat{\beta}$ rounded to three decimals):**

$$\lambda(t | X) = \lambda_0 e^{0.630 * hormone - 0.015 * age - 0.393 * menopausal + 0.007 * size + 0.083 * nodes - 0.038 * nodes * hormone}$$

Model 3 output: exponentiated coefficients

Variable	Hazard Ratio	Std Error	Z statistic	p-value	Lower 95% CI	Upper 95% CI
Hormone: Yes	1.878	0.167	3.766	0.000	1.353	2.607
Nodes involved	1.086	0.014	5.991	0.000	1.057	1.116
Age (years)	0.985	0.009	-1.674	0.094	0.968	1.003
Menopausal: Yes	0.675	0.179	-2.195	0.028	0.475	0.959
Tumor size (mm)	1.007	0.004	1.666	0.096	0.999	1.014
Hormone: Yes $\times$ nodes	0.963	0.015	-2.480	0.013	0.934	0.992

Model 3 extends Model 2 by including an interaction between hormone therapy and number of involved lymph nodes. The main effect of hormone therapy (HR = 1.878, 95% CI: 1.353–2.607, $p = 0.000$) is the hazard ratio when `nodes` = 0. The interaction term is statistically significant (HR = 0.963, 95% CI: 0.934–0.992, $p = 0.013$) at $\alpha = 0.05$. This suggests that the effect of hormone therapy on recurrence depends on the number of nodes involved.

The interaction hazard ratio of 0.963 suggests that for each additional involved node, the hazard ratio for hormone therapy decreases by approximately 3.7%. In other words, the effect of hormone therapy on the hazard of recurrence becomes smaller as the number of nodes increases.